

The Human Eye and the Colourful World

Science • Chapter 10 • Chapter Summary & Study Guide for Daksh

Your eye is a remarkable optical instrument — it can adjust focus to read a book in your hand and almost instantly switch to a distant mountain, work in dim light or bright sunshine, and detect millions of colours. This chapter explains the working of the human eye, common defects of vision and their corrections, and everyday optical phenomena: dispersion of white light, atmospheric refraction, scattering — including the blue sky and red sunset.

How the human eye works

Light passes through the **cornea** (which does most of the focusing), the **aqueous humour**, then the **pupil** (whose size is regulated by the iris muscles depending on light intensity), the **convex eye-lens**, and the **vitreous humour**, finally forming a real, inverted image on the **retina**. Photoreceptors there (rods for low light, cones for bright light + colour) convert light into nerve impulses, sent through the **optic nerve** to the brain — which interprets them right-side-up.

The eye-lens is held by **ciliary muscles**. By contracting/relaxing, these muscles change the lens's thickness, hence its focal length — letting us focus on near or far objects. This ability is called **power of accommodation**. The closest distance at which the eye can see comfortably is the **least distance of distinct vision**, ~25 cm for a normal adult.

Defects of vision and their correction

- **Myopia** (short-sightedness) — can see near, can't see far. Image forms in front of the retina because either the eyeball is elongated or the lens is too curved. *Corrected with a concave (diverging) lens.*
- **Hypermetropia** (long-sightedness) — can see far, can't see near. Image forms behind the retina because the eyeball is too short or the lens is too flat. *Corrected with a convex (converging) lens.*
- **Presbyopia** — age-related weakening of the ciliary muscles → loss of accommodation; old people often need **bifocal lenses** (concave on top for distance, convex below for reading).
- **Cataract** — clouding of the lens; treated by surgical replacement with an artificial lens.

Refraction through a prism — dispersion

When white light passes through a glass prism, it splits into a band of seven colours: **VIBGYOR** (Violet, Indigo, Blue, Green, Yellow, Orange, Red). This is called **dispersion**. Different colours have different wavelengths and hence different refractive indices in glass — violet bends most, red least. Newton showed that white light is a mixture of these colours by recombining the spectrum with a second prism.

A **rainbow** is a natural dispersion + reflection effect — water droplets act like millions of tiny prisms after a rain shower.

Atmospheric refraction

Air at different heights has different densities, so light bends as it passes through layers of varying density. Effects:

- **Twinkling of stars** — apparent position of a star keeps shifting due to changing refraction; planets are nearer and don't twinkle.
- **Apparent flattening of sun** at sunrise/sunset and the sun being visible 2 minutes before actual rise / after actual set — atmospheric refraction.
- **Mirage** in deserts — hot air near ground bends light, giving an illusion of water.

Scattering of light

Tyndall effect: when light passes through a medium with fine suspended particles, it gets scattered, making the path visible — observed in mist, smoke, or a beam through a forest canopy.

Why is the sky blue?

Air molecules scatter shorter (blue) wavelengths much more strongly than longer (red) ones. So light reaching our eyes from the sky is predominantly blue.

Why is sunset/sunrise red?

Sunlight passes through a much thicker layer of atmosphere -> blue is scattered out, only red/orange reaches us. Same reason 'danger' signals are red - red light penetrates fog/dust the best.

★ MUST-REMEMBER POINTS

- Image on retina is REAL and INVERTED; brain flips it upright.
- Power of accommodation = ability of eye lens to change focal length.
- Myopia (–ve lens), Hypermetropia (+ve lens), Presbyopia (bifocal), Cataract (surgery).
- Least distance of distinct vision ≈ 25 cm. Far point of normal eye = infinity.
- VIBGYOR: violet bends most, red least in a prism.
- Sky is blue due to scattering (shorter λ scatters more); sunset is red because longer path $\rightarrow$ only long λ reaches eye.
- Stars twinkle (atmospheric refraction); planets don't (much closer).

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